

Ionic and Covalent Bonding

Coulomb's law – the attraction and repulsion between charges – affects the properties of atoms, and is responsible for the phenomenon of chemical bonding.

1. Atoms share electron pairs between them. These relationships are called chemical bonds.
2. Atoms have a property called electronegativity, which is a measure of how much they attract electrons toward themselves.

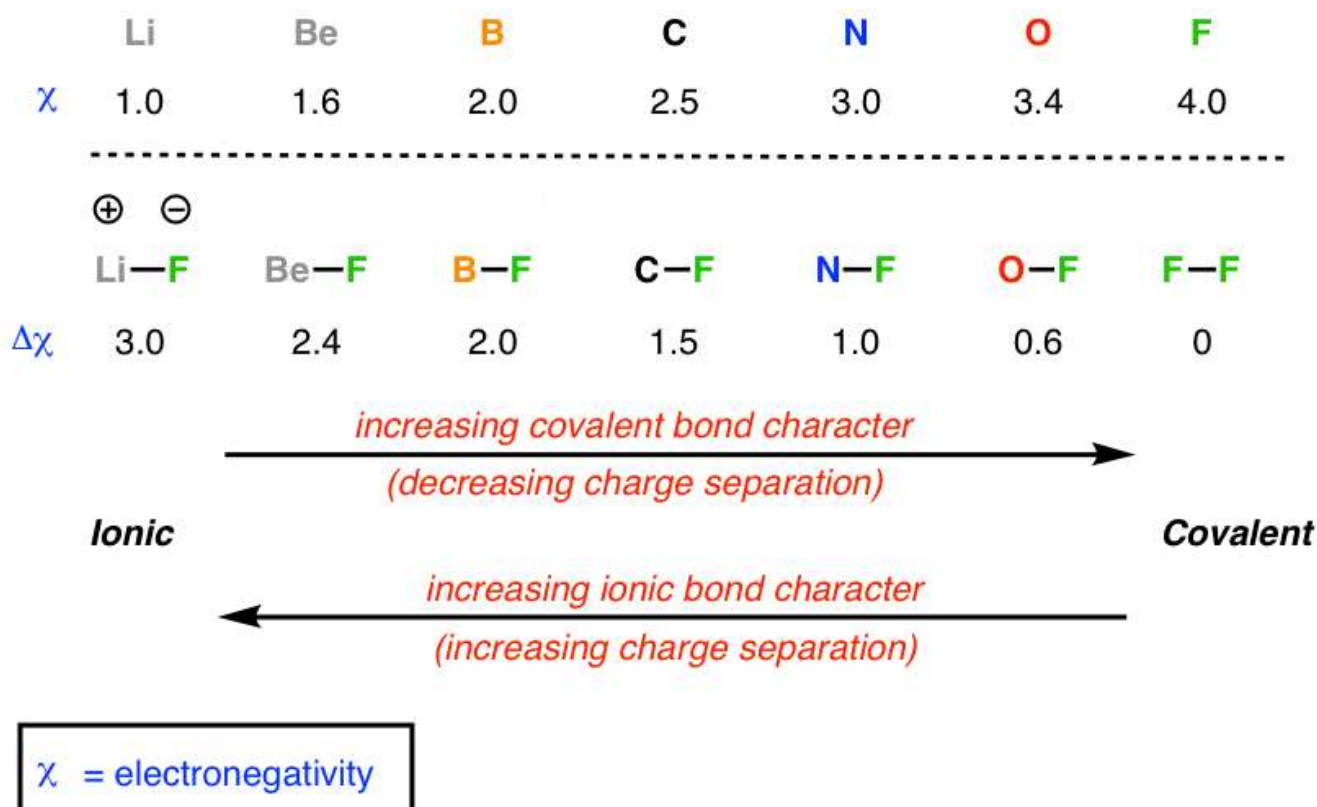
Here's the important consequence of these two facts: when bonds form between atoms of different electronegativity, the electron pair in the bond will not be shared equally. It will be **polarized** toward the atom with the greater electronegativity. What this means is that the more electronegative element will have greater electron density (a net negative charge) and the less electronegative element will have lower electron density (a net positive charge).

The greater the difference in electronegativity, the greater the polarization.

Sometimes the "sharing" of electrons to form a bond is sharing in name only. Let's say the parents of two boys, aged 10 and 8, give them a \$20 bill "to share" between them, and walk away. Assuming no further parental interference, what do you think the odds are that the money will get split 50/50? Pretty slim (I speak from personal experience). Although the two brothers are *nominally* sharing \$20, it's the older brother who has the power to deliver an unbreakable Full Nelson, and thus holds the majority of the purchasing power.

So it is with atoms. At one end of the scale you have the bonds of a highly electronegative element like fluorine with a poorly electronegative element like lithium. In lithium fluoride, the bond is so highly polarized that essentially one can think of lithium's electron as residing exclusively on the fluorine, so that fluorine has a full octet. This type of bonding is referred to as ionic [ions = charged atoms or molecules], and the bonding behaves essentially like the attraction between two point charges obeying Coulomb's law.

On the other end of the scale, you can have an element like fluorine bound to itself, where there is no dipole (electronegativity difference being zero) and hence the bonding is not ionic, but what we call covalent – a full and equal partnership.



In between the extremes there are shades of both. Here's an important point. Just as one should be hesitant in looking at issues in black and white, chemical bonding is no different. There's a temptation to look at bonds as either covalent OR ionic; instead, it may be more helpful to look at in terms of flavor or character – bonds can have differing amounts of covalent or ionic *character*. So the middle cases have a lot of shades of both.

So what implications does this have for organic chemistry? Too many to talk about in depth! **This is the key phenomenon which underlies a lot of the richness of chemistry.** But here are a few examples.

- **Boiling points/melting points** – *in general*, the more polarized the molecule, the higher the boiling point/melting point. Water is a highly polarized molecule, considering the difference in electronegativity between oxygen (3.4) and hydrogen (2.2) reflected in its extremely high boiling point (relative to its molecular weight) of 100 °C.
- **Solubility** – You’ve probably come across this as the easy-to-remember slogan “like dissolves like”. *In general*, the more polarized the molecule, the better will be the solubility in a polar solvent (like water, with its polarized O-H bonds).
- **Acidity** – the electronegativity of the atom bound to hydrogen is an important factor (though not the only one!) in determining the acidity of a molecule
- **Reactivity** – for organic chemistry, this is extremely important for the purposes of looking at carbon. Atoms with high electron density will tend to form bonds at carbons with low electron density, and vice versa.

Learning to recognize dipoles at carbon is one of the key skills that will help you determine its reactivity, as you’ll see when you learn about electrophilicity (electron-deficiency) and nucleophilicity (electron-richness).

Analogies

I often think of electronegativity as one of the key factors that give atoms their unique personalities. You can think of electronegativity as being a lot like greed, with fluorine being the greediest of all, and cesium as a big absent-minded oaf that loses track of its only electron all the time.

This is a brief treatment of one of the most fundamental concepts in chemistry, but we’ll be coming back to it again and again. **Take home message:** *Pay close attention to the differences in electronegativity between atoms!*

(Advanced) References And Further Reading

1. **Principles of electronegativity Part I. General nature.** R. T. Sanderson *Journal of Chemical Education* **1988** 65 (2), 112
DOI: [10.1021/ed065p112](https://doi.org/10.1021/ed065p112)
2. **Principles of electronegativity Part II. Applications.** R. T. Sanderson *Journal of Chemical Education* 1988 65 (3), 227
DOI: [10.1021/ed065p227](https://doi.org/10.1021/ed065p227)

These two papers by R.T. Sanderson of Arizona State University has a brief and very clear discussion of the concept of electronegativity, how it has been historically measured, and how it may be improved, as well as a discussion on the importance of *partial charges* in aiding student understanding of chemical reactions.